DES423 · PREDICTIVE MODELING PITCH

SessionSignal

Can browsing behavior tell us which shopping sessions will end in a purchase?

UCI Online Shoppers Purchasing Intention · 12,330 sessions · live demonstration · 7-member team

PROBLEM IMPORTANCE

Lost sessions are a large, persistent cost

$6.42T

global retail ecommerce sales in 2025 — 20.5% of all retail

70.22%

average documented cart abandonment, 50 studies 2006–2025

84.5%

of sessions in our dataset ended without a purchase

Abandonment has sat near 70% for two decades, so this is a **structural gap, not a temporary dip**. At a $6.42 trillion market scale, even small conversion gains are commercially significant.

No store can hand-review every lost session. **Analyst attention is the scarce resource** — and that is what we target.

Sources: eMarketer 2025 forecast; Baymard Institute (unweighted average of 50 studies, range 55%–84%); UCI dataset 468. Cart abandonment and session non-purchase are related but different measures.

PROBLEM STATEMENT

Which lost sessions deserve attention first?

User: an ecommerce conversion analyst.

Situation: thousands of sessions end without a purchase; only a handful can ever be investigated.

Question: which ones should the team examine first?

Objective: rank completed sessions by purchase probability, using browsing behavior alone.

How we measure success

Concentrate real buyers into the top-scored slice. Our primary metric is **average precision**, because only 15.5% of sessions convert — plain accuracy would reward always predicting “no purchase”.

Explicitly out of scope

Why a shopper left · guaranteed revenue recovery · real-time intervention mid-session.

DATASET AND TARGET

One row = one shopping session

12,330
historical sessions

17
input features in the original data

15.5%
ended in a purchase

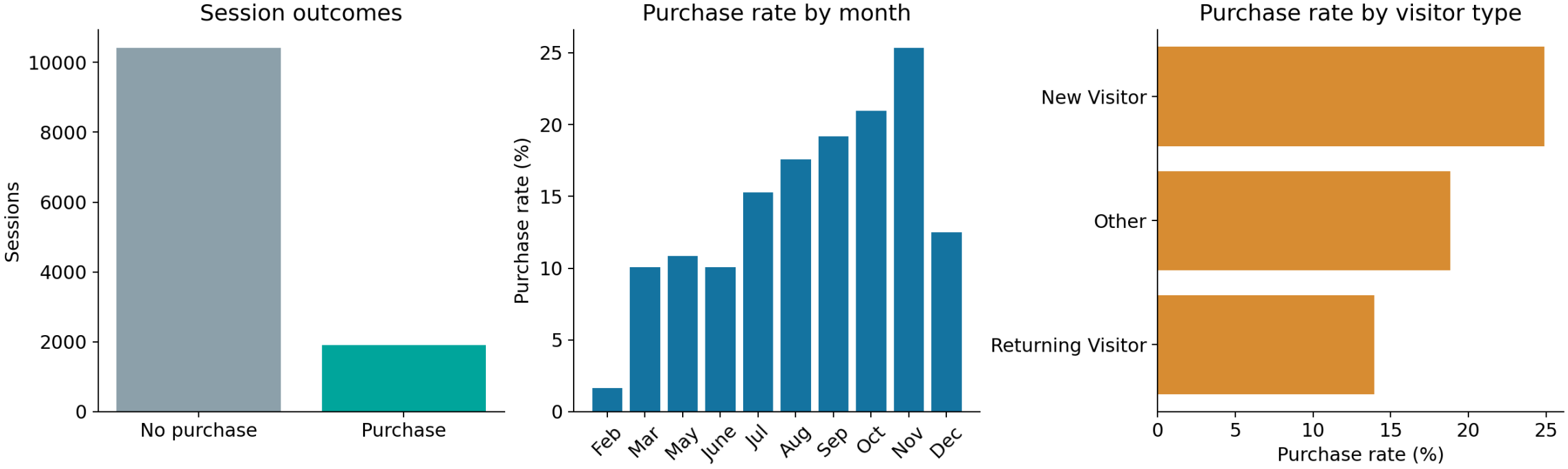
Predict: Revenue = purchase or no purchase.

Inputs: page counts and time, analytics rates, visitor type, month, device and traffic categories.

Source: UCI Machine Learning Repository, dataset 468

EXPLORATORY ANALYSIS

Purchases vary by month and visitor type

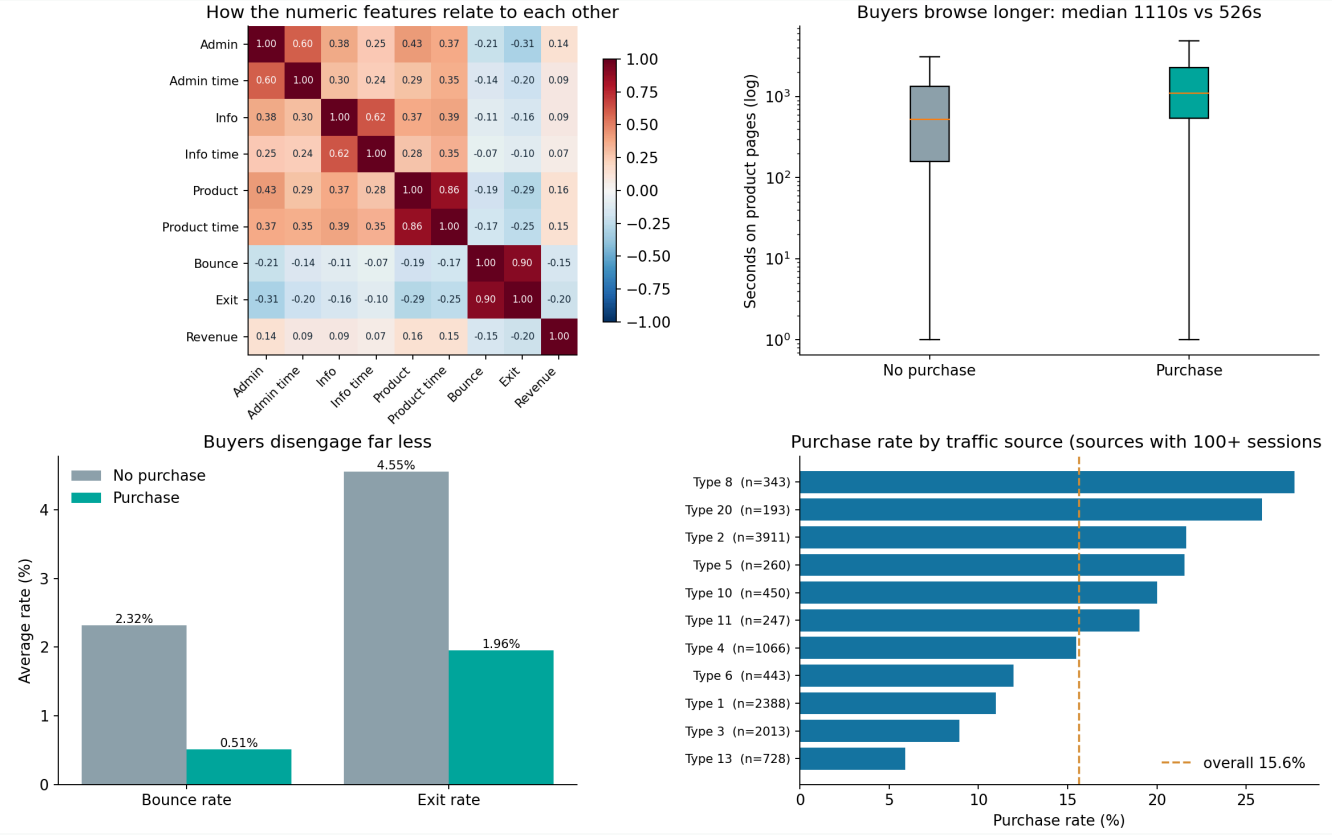


New visitors purchased in 24.9% of sessions versus 13.9% for returning visitors. November reached 25.4%; February was 1.6%.

Calculated from the supplied UCI CSV. These are associations, not causal effects. Month groups have different sample sizes.

EXPLORATORY ANALYSIS

Buyers behave measurably differently



Buyers spend about **twice as long** on product pages (median 1,110s vs 526s) and disengage far less (exit rate **1.96% vs 4.55%**). Exit rate is the numeric feature most related to buying (r = −0.20); bounce and exit move together (r = 0.90). Traffic source 8 converts at 27%, source 13 at 6%.

Computed from the 12,205 cleaned sessions. Associations, not causes. Traffic sources shown have 100+ sessions.

EVALUATION DESIGN

Keep the test set unseen until the end

01 Remove 125 exact duplicate rows

02 Exclude `PageValues`, a revenue-derived analytics feature

03 Stratified 80/20 split: 9,764 train and 2,441 test sessions

04 Four-fold cross-validation and tuning on training data only

Primary measure: average precision (AP), because purchases are uncommon.

We measured what step 02 costs: keeping `PageValues` would lift AP from **0.374 to 0.723**. It is computed from revenue, so we treat that as **leakage, not an achievement**.

MODELS

Models for purchase classification

Logistic regression

Combines feature values into a linear score. Class weighting gives rare purchases more influence during training.

Random forest

Averages predictions from many decision trees to capture nonlinear patterns and interactions.

Gradient boosting

Adds trees sequentially to improve earlier predictions. The main pipeline uses histogram gradient boosting.

No-purchase baseline

Predicts no purchase for every session. Its high accuracy shows why purchase recall also matters.

PYCARET CLASSIFICATION

The same question, asked across 14 models

Low-code workflow

1. `setup` — 80/20 split, four folds, `fix_imbalance=True`
2. `compare_models` — PyCaret's full classifier list
3. `tune_model` → `predict_model` on its own holdout

Same cleaned data, same excluded `PageValues`.

Top of the leaderboard (CV AUC)

Model	AUC	Recall
LightGBM	0.7732	14.9%
Gradient boosting	0.7641	12.8%
Random forest	0.7495	12.1%
Dummy baseline	0.5000	0.0%

Best PyCaret AUC **0.773** against our scikit-learn **0.787**. Two toolchains, one conclusion: **no model pulls clearly ahead.**

Saved run: full 14-model leaderboard. PyCaret uses its own split and preprocessing, so read it alongside — not merged into — the main results.

HELD-OUT RESULTS

Read the intervals, not the ranking



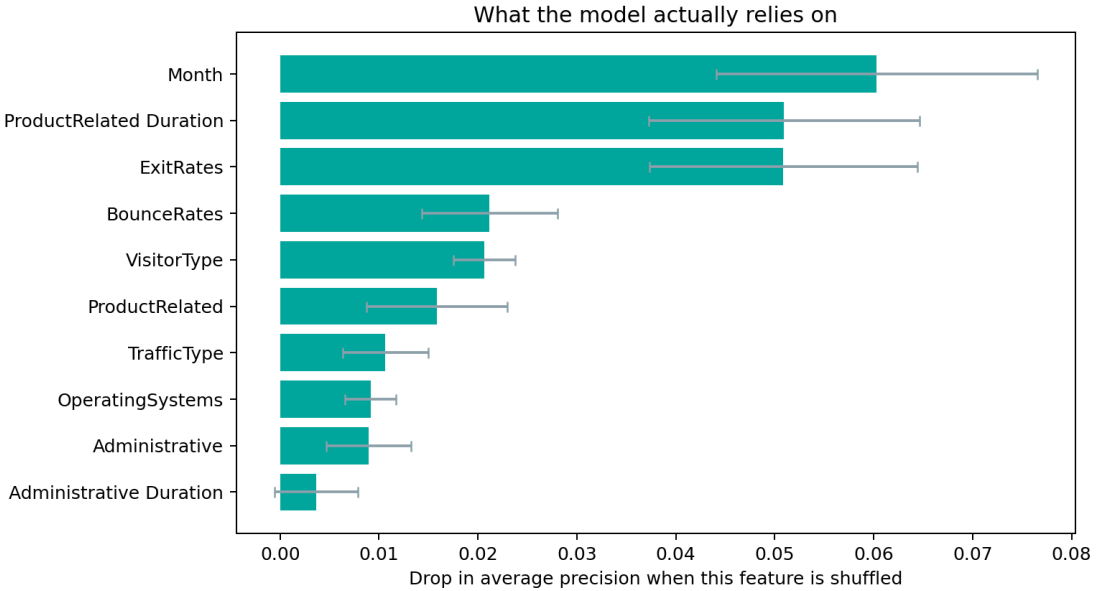
The tuned forest leads by **+0.0028** — 95% CI **-0.008 to +0.014**, winning only **68%** of 2,000 bootstrap resamples. The interval crosses zero.

So we do not claim a winner. Every real model beats the baseline decisively; between themselves they are statistically indistinguishable.

Average precision on the same untouched 2,441-session test set; 95% intervals from 2,000 bootstrap resamples.

INTERPRETATION

What is the model actually using?



Drop in AP when shuffled

Feature	Impact
Month	−0.060
ProductRelated Duration	−0.051
ExitRates	−0.051
BounceRates	−0.021
VisitorType	−0.021

The top feature is `Month` — not a behaviour. The model has partly learned the seasonality of one store in one year.

That is a limitation we found by checking, and it is exactly why validation on a later time period is our first next step.

Permutation importance on the held-out set, 10 repeats, measured in average precision.

IMPROVEMENT

The threshold mattered more than the hyperparameters

Operating point	Precision	Recall	F1
0.500 — the default	37.1%	46.3%	0.412
0.446 — best F1	**34.5%**	**62.3%**	**0.444**
0.607 — review top 10%	43.7%	28.0%	—

Grid-searching hyperparameters moved AP by **+0.002** — noise. Moving the decision threshold lifted recall from **46.3%** to **62.3%** and F1 from **0.412** to **0.444**, for free.

Thresholds were selected on the test set and would need a separate validation split before operational use.

SOLUTION

From a score to a worklist

1. Score every completed session.
2. Rank them and take as many as the team can review.
3. Pull out the high scorers that **did not** buy.
4. Investigate those, form one hypothesis, A/B test it.

Measured on the holdout

25

real buyers found in the top **50** sessions, against **8** expected at random — **3.2×**

Those 50 also contain **25 high-intent sessions that bought nothing.** That list is the product.

Steps 1–3 run in the app today. Step 4 needs journey logs and an experiment.

From `artifacts/scored_sessions.csv`: all 2,441 held-out sessions, scored once by the chosen model.

LIVE DEMONSTRATION

Two minutes, two screens

```
Score a session — held-out visitors the model never saw
  #27 → 68% · bought  ✓   #9 → 1% · did not buy  ✓
  #8 → 64% · did not buy  ← investigate

Find lost sales — rank all 2,441 and open the worklist
```

Visitor #8 browsed **38 product pages over 64 minutes** and never bounced — then left with nothing. That is the session an analyst should open first.

NEXT STEPS

Make the model more trustworthy

- Collect timestamped partial sessions for genuine live prediction.
- Validate on a later time period and another store.
- Check calibration, subgroup performance, and false positives.
- Choose an action threshold based on cost and capacity.
- Run an A/B test before claiming sales impact.

TEAM AND CONTRIBUTIONS

Who did what

Student ID	Name	Main contribution
6622782298	Sirawich Narakulsookpipat	Problem research and framing
6622780342	Chaitawat Boonkitticharoen	Dataset audit and target definition
6622782215	Bhuchi Kumprakob	Exploratory data analysis
6622780276	Nonthaphat Chuathong	Evaluation design and model selection
6622770657	Kamin Sirisuwapong	PyCaret comparison and hyperparameter tuning
6622772455	Poonnawit Wongprateepwilai	Results analysis and business solution
6622790267	Choltawat Jullakasevi	Streamlit app, demonstration and limitations

Every member presents their own section.

TAKEAWAY

We can rank purchase intent. We still need to test the intervention.

The tuned forest improves AP from 0.156 to 0.374 on held-out sessions.

Sirawich · Chaitawat · Bhuchi · Nonthaphat · Kamin · Poonnawit · Cholwat